

Declan Chan

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Education

BASc in Engineering Science

2025–Present

University of Toronto

- Dean's Honours List (every semester)

IB Diploma Programme

2021–2025

Maple High School

- IB Score: 45/45
- OSSD Top 6: 99.67%
- Director's Achievement Award

Technical Skills

Languages: Python, C, MATLAB, Bash, SQL

Design: SolidWorks, OnShape

Tools: Git, Linux, Slurm

Experience

Research Fellow

2026–Present

Autonomous Discovery of Alloys

- Designed experiments to evaluate the robustness of machine learning force fields.
- Benchmarked model predictions against density functional theory relaxations.
- Performed large-scale experiments using Canada's Fir HPC cluster.
- Presented findings at Undergraduate Engineering Research Day Conference.

Space Systems Engineer

2026–Present

University of Toronto Aerospace Team

- Developed software for CubeSat attitude determination and control.
- Built Python modules for attitude dynamics and flight simulation.
- Simulated magnetic environments using a Helmholtz cage and MATLAB.

Mechanical Design Engineer

2025–2026

Robotics for Space Exploration

- Designed and modelled robotic arm components using 3D CAD.
- Performed finite element analysis in SolidWorks Simulation to evaluate designs.

Senior Instructor

2021–Present

Northern Karate Schools

- Yondan: Certified 4th-degree Black Belt.
- Gold medal at the Toronto International Tournament of Martial Arts Champions.
- Taught 100+ students each week through group and private lessons.

Projects

Machine Learning Force Field Attacks

2026

- Developed open-source software to generate gradient-based adversarial attacks against MLFFs.
- Built Python/PyTorch pipelines to process Materials Project and Li-C-O datasets.
- Automated ASE relaxations, perturbations, and contour exploration workflows using Bash and Slurm.

Neural Lab

2026

- Developed a configurable PyTorch image-classification pipeline supporting reproducible training.
- Implemented workflows for MNIST, Fashion-MNIST, CIFAR-10, and SVHN.
- Automated experiment tracking with SQLite and SQL to compare accuracy of model configurations.

Medical RAG Platform

2026

- Developed an interactive platform visualizing human disease progression.
- Implemented a retrieval-augmented generation system using PubMed data via IBM watsonx.ai.
- Built an AI-powered 3D web application with React and Three.js.

Bridge Design Optimization

2025

- Optimized structural performance and prototyped a matboard bridge using Python.
- Ranked 1st in the cohort and 2nd in the Class of 2025 for CIV102.

Generative Art Engine

2021–2022

- Used the HashLips Art Engine to generate layered NFT artwork from image assets.
- Created and deployed NFTs on OpenSea.io using the Polygon blockchain.

Extracurricular Achievements

Speedcubing Competitions

2022–2025

- Ranked 90th in Canada for 3x3x3 fewest moves.
- Judged participants at the World Cube Association.
- Club President, taught beginners and organized school-wide competitions.

Varsity Volleyball

2022–2025

- Team Captain in Grades 10 and 12.
- 2nd place, Tier 1 YRAA Championship junior varsity.
- 2nd place, Tier 1 YRAA Championship senior varsity.

Rep Volleyball

2019–2023

- Team Captain and 2× MVP.
- Silver medal at Canada's 2023 Youth Nationals.

Music Performance

2019–2023

- Level 10 RCM Theory and Instrumental Comprehensive Certificate (First Class Honors).
- Level 10 RCM Practical Piano Certificate (First Class Honors with Distinction).
- Level 8 RCM Practical Violin Certificate (First Class Honors with Distinction).